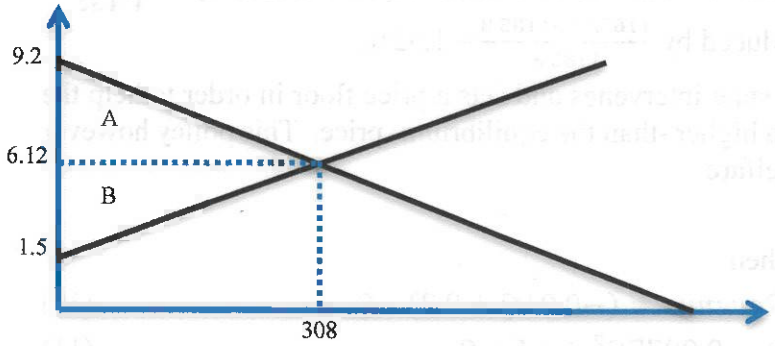
Power System Economics – Answers

Solution to Question 1:

(a) First of all, electricity is inextricably linked with a physical delivery system which operates much faster than any market. Generation and load must be balanced at all times because failure to balance leads to collapse of the system and blackouts, which have enormous economic consequences. As a result, physical balance of the system cannot be left entirely to a market.
Secondly, electricity cannot be stored in large quantities as most storage technologies are characterised by high capital costs and low efficiency. As a result, the total production needs to be equal to the total consumption at any given time.
Thirdly, the demand curve is almost vertical, meaning that the price elasticity of the demand is very low. In other words, the consumers do not often respond to high market prices, making the exercise of market power by large generators more probable.
Finally, electricity is inextricably linked with a network characterised by certain physical laws. Therefore, the electricity produced by different generators and electricity consumed by different consumers is indistinguishable and thus pooled. In other words, a generator cannot physically direct its produced electricity to a specific consumer and a consumer cannot choose which specific generator produces its load.

(b) The *supply curve* indicates the value that the market price should take to make it worthwhile for the aggregated producers to supply a certain quantity of the commodity to the market. Similarly, the *demand curve* describes the quantity of products that the aggregated consumers are willing to purchase for a particular price in a given market. Demand curve is normally downward sloping, while the supply curve is normally upward sloping. *Market clearing price* or the *equilibrium price* is the price for which the quantity that the suppliers are willing to provide is equal to the quantity that the consumers wish to obtain. *Social welfare* is the sum of the net consumers' surplus and of the producers' profit, and it quantifies the overall benefit that arises from trading.

(c)



(c) (i) Inverse demand function:

$\pi_D = -0.01Q_D + 9.2$ (1)

Marginal generation cost = inverse supply function:

$\pi_S = 0.015Q_S + 1.5$ (2)

where Q_D, Q_S in kW and π_D, π_S in £/kWh.

Equilibrium is located at the point:

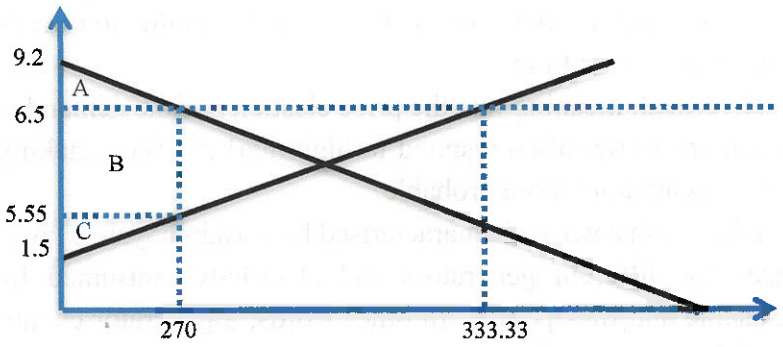
$$\pi_s = \pi_D \rightarrow Q_S = Q_D = Q^* = 308kW, \quad \pi^* = 6.12 \frac{£}{kWh} \tag{3}$$

Consumer surplus equal to the area of A = 0.5 × 308 × (9.2 - 6.12) = 474.32£/h (4)

Producer surplus equal to the area of B = 0.5 × 308 × (6.12 – 1.5) = 711.48£/h (5)

Total social welfare equal to the area of A+B = 474.32 + 711.48 = 1,185.8£/h (6)

(c) (ii)



In this case, the market does not operate at equilibrium because $\pi_D = 6.5 \frac{£}{kWh}$. For this price, the demand is 270kW and the supply -theoretically- is 333.33kW. However, because electricity supply must equal demand, the supply is 270kW as well. If it was a different commodity, then the extra capacity could be bought by the government and stored for future potential situations of shortage. But this cannot be done with electricity.

Consumer surplus equal to the area of A = 0.5 × 270 × (9.2 - 6.5) = 364.5£/h (7)

Producer surplus equal to the area of B+C = 270 × (6.5 – 5.5) + 0.5 × 270 × (5.55 – 1.5) = 803.25£/h (8)

Total social welfare equal to the area of A+B+C = 364.5 + 803.25 = 1,167.75£/h (9)

Compared to the equilibrium result obtained in (i), it can be observed that the consumption is decreased by $\frac{270-308}{308} = 12.34\%$, the price is increased by $\frac{6.5-6.12}{6.12} = 6.21\%$, and the producer surplus increased by $\frac{803.25-711.48}{711.48} = 12.90\%$ and the consumer surplus reduced by $\frac{364.5-474.32}{474.32}$ = 23.15%, the total social welfare reduced by $\frac{1167.75-1185.8}{1185.8} = 1.52\%$.

This is a case when for example the state intervenes and sets a price floor in order to help the suppliers by allowing them to sell at a higher -than the equilibrium- price. This policy however results in a reduction of the social welfare.

(iii) The price $\pi = -0.01Q + 9.2$, then

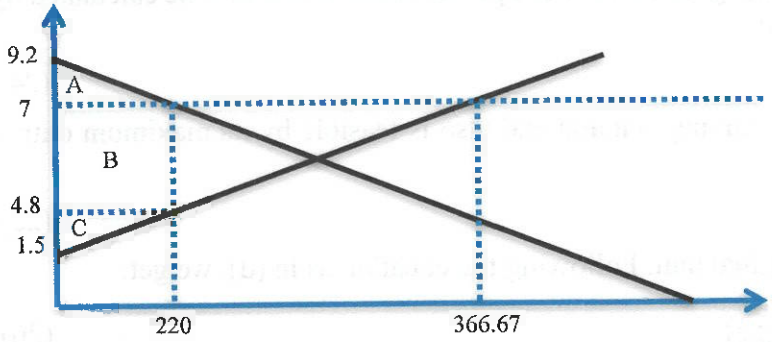
The producer's revenue function is $Revenue = (-0.01Q + 9.2) * Q$ (10)

The producer's cost function is $Cost = 0.0075Q^2 + 1.5 * Q$ (11)

The producer's profit function is $Profit = -0.0175Q^2 + 7.7 * Q$ (12)

In order to maximize producer's profit, we set $\frac{\partial profit}{\partial Q} = -0.035 * Q + 7.7 = 0$ (13)

Then, the production and price at the profit maximization point is $Q = 220$kW, $\pi = 7$£/kWh.



Therefore, to maximize its profit, this producer needs to artificially increase the price by $\frac{7-6.12}{6.12} = 14.38\%$.

No, it will not be worth for the producer to further increase or decrease the price, since further increasing the price will leads to a decreased quantity sold by the producer, the increase in price however does not off-set the reduction in quantity. Similarly, further decreasing the price will leads to an increased quantity sold by the producer, this increase in quantity however does not off-set the reduction in price. For this producer, the optimal trade-off of higher price and lower quantity is achieved at the profit maximization point.

(d) In the optimal solution of the problem, the marginal cost of each generator is equal to the system marginal cost λ yielding:

$$\frac{\partial C_1(P_1)}{\partial P_1} = 8.05 + 0.4 * P_1 = \lambda \rightarrow P_1 = \frac{\lambda - 8.05}{0.4} \quad (14)$$

$$\frac{\partial C_2(P_2)}{\partial P_2} = 8.45 + 0.6 * P_2 = \lambda \rightarrow P_2 = \frac{\lambda - 8.45}{0.6} \quad (15)$$

The satisfaction of the system demand implies that:

$$P_1 + P_2 = D = 600\ [MW] \quad (16)$$

Substituting (14)-(15) in (16) gives for the system marginal cost:

$$\frac{\lambda - 8.05}{0.4} + \frac{\lambda - 8.45}{0.6} = 600 \rightarrow \lambda = 152.21\ \left[\frac{£}{MWh}\right] \quad (17)$$

Substituting (17) in (14)-(15) yields for the optimal production of each generator:

$$P_1 = 360.40\ [MW] \quad (18)$$

$$P_2 = 239.60\ [MW] \quad (19)$$

Substituting (18)-(19) in the cost functions of the generators yields for the total cost of operating the system:

$$C_{tot} = C_1(P_1) + C_2(P_2) = 48{,}990.92\ [£/h] \quad (20)$$

The system average cost is:

$$\alpha = \frac{C_{tot}}{P_1 + P_2} = 81.65\ \left[\frac{£}{MWh}\right] \quad (21)$$

The profit for each generator Ω_i is given by the difference between its respective revenue and cost:

$$\Omega_1 = \lambda * P_1 - C_1(P_1) = 25{,}677.63\ [£/h] \quad (22)$$

$$\Omega_2 = \lambda * P_2 - C_2(P_2) = 16{,}657.45\ [£/h] \quad (23)$$

(e) Given that $P_1 > P_1^{max}$ the solution calculated in **(d)** is not feasible when taking into account given maximum output limits of the generators. The optimal solution can now be calculated by fixing the output of generator 1 to:

$$P_1' = P_1^{max} = 350\ [MW] \tag{24}$$

and generator 2 satisfying the remaining demand and also is feasible by its maximum output limit 300MW:

$$P_2' = D - P_1' = 250\ [MW] \tag{25}$$

Thus, the generators 2 is the marginal unit. Following the equation as in (d), we get:

$$\frac{\lambda' - 8.45}{0.6} = 250 \rightarrow \lambda' = 158.45\ [\frac{£}{MWh}] \tag{26}$$

Substituting (24) and (25) in the cost functions of the generators yields for the total cost of operating the system:

$$C_{tot}' = C_1(P_1') + C_2(P_2') = 49{,}045.00\ [£/h] \tag{27}$$

The system average cost is:

$$a' = \frac{C_{tot}'}{P_1' + P_2'} = 81.47\ [\frac{£}{MWh}] \tag{28}$$

The profit for each generator Ω_i is given by the difference between its respective revenue and cost:

$$\Omega_1' = \lambda * P_1' - C_1(P_1') = 27{,}840.00\ [£/h] \tag{29}$$

$$\Omega_2' = \lambda * P_2' - C_2(P_2') = 18{,}185.00\ [£/h] \tag{30}$$

When the maximum output limits are considered, the output of generator 1 needs to be restricted to a lower value and therefore the output of the generator 2 is increased. Given that generator 2 is characterised by higher variable cost, this new dispatch is more costly (i.e. it increases the total cost of operating the system, the system marginal cost and the average system cost). On the other hand, this new dispatch increases the profit of both the generators. For generators 2 this is driven by the increase of both its outputs and the system marginal price, while for generator 1 this is driven by the increase of the system marginal price despite the reduction of its output.

Solution to Question 2:

Discussion along the following line is needed for full mark.

(a) Unbundling refers to the process of separating the activities of generation, transmission, distribution and retail of electricity, previously performed by a single vertically integrated utility, into different entities i.e. companies. This then allows for the competition to be introduced in the generation and retail business, while transmission and distribution are organised as regulated monopolies, ensuring fair and open network access to all market participants. The reasons for liberalising electricity markets are the increased efficiency in the supply of electricity and the lower cost of electricity to consumers. This is achieved through introducing competition both at the wholesale level (generators compete to sell electrical energy) and the retail level (consumers choose from whom they buy electricity).

If a generator bids higher than its marginal cost, it could become an extra-marginal producer and miss an opportunity to sell at a profit. If the generator bids lower than its marginal cost, it could have produced at a loss. Any of these two options bring in a certain amount of risk to the

producer that would rather be avoided. On the other hand, if the producer bids at its marginal cost, it gets paid the market price if it is a marginal or infra-marginal producer. This is the preferred option for a producer.

(b) Under perfect competition, the price is equal to the marginal cost of each generator. Therefore, the following two equations hold:

$$\frac{\partial C_A(P_A)}{\partial P_A} = \pi \quad (1)$$

$$\frac{\partial C_B(P_B)}{\partial P_B} = \pi \quad (2)$$

Since $\pi = 125 - 1.7 \cdot D$ and $D = P_A + P_B$ (demand-supply balance condition), equations (1) and (2) yield respectively:

$$35 + 0.74 \cdot P_A = 125 - 1.7 \cdot (P_A + P_B) \rightarrow P_A = 36.8852 - 0.6967 \cdot P_B \quad (3)$$

$$39 + 0.5 \cdot P_B = 125 - 1.7 \cdot (P_A + P_B) \rightarrow P_B = 39.0909 - 0.7727 \cdot P_A \quad (4)$$

The combination of (3) and (4) yields:

$$P_A = 20.9040\ [MW] \quad (5)$$

$$P_B = 22.9379\ [MW] \quad (6)$$

The total demand is given by:

$$D = P_A + P_B \rightarrow D = 43.8418\ [MW] \quad (7)$$

The electricity price is given by:

$$\pi = 125 - 1.7 \cdot D \rightarrow \pi = 50.4689\ [\tfrac{£}{MWh}] \quad (8)$$

The profits made by the two generators are given by:

$$\Omega_A = P_A \cdot \pi - 35 \cdot P_A - 0.37 \cdot P_A^2 \rightarrow \Omega_A = 161.6809\ [\tfrac{£}{h}] \quad (9)$$

$$\Omega_B = P_B \cdot \pi - 39 \cdot P_B - 0.25 \cdot P_B^2 \rightarrow \Omega_B = 131.5363\ [\tfrac{£}{h}] \quad (10)$$

(c) (i) In the Cournot model of competition the state of the market is determined by the production decisions made by each firm. We summarize the possible outcomes using a table where all the cells in a column correspond to a given production by company A and the cells in a row correspond to a given production by company B.

Each cell contains four pieces of information arranged in the following format:

D	Ω_A
Ω_B	π

where:

π price $[\tfrac{£}{MWh}]$

D demand $[MW]$

Ω_A profit made by company A $[\frac{£}{h}]$

Ω_B profit made by company B $[\frac{£}{h}]$

Given the productions P_A and P_B of the two companies, the other quantities are calculated as follows:

$$D = P_A + P_B \tag{11}$$

$$\pi = 125 - 1.7 \cdot D \tag{12}$$

$$\Omega_A = P_A \cdot \pi - 35 \cdot P_A - 0.37 \cdot P_A^2 \tag{13}$$

$$\Omega_B = P_B \cdot \pi - 39 \cdot P_B - 0.25 \cdot P_B^2 \tag{14}$$

Based on the above, the table expressing the Cournot model of competition for the conditions of the problem is shown below.

P_B/P_A	14		15		16	
15	29	497.28	30	501.75	31	502.08
	494.25	75.7	468.75	74.0	443.25	72.3
16	30	473.48	31	476.25	32	474.88
	496.00	74.0	468.80	72.3	441.60	70.6
17	31	449.68	32	450.75	33	447.68
	493.85	72.3	464.95	70.6	436.05	68.9

A Nash equilibrium state implies that given the production level of either of the two companies in this state, the other company cannot achieve a higher profit by following a production level different than the one at the equilibrium. It can thus be observed form the table below that the cell corresponding to $P_A = 16MW$ and $P_B = 15MW$; $P_A = 15MW$ and $P_B = 16MW$ are two Nash equilibria. In other words, given that $P_A = 16MW$, company B achieves its highest profit by producing $P_B = 15MW$ AND given that $P_B = 15MW$, company A achieves its highest profit by producing $P_A = 16MW$. Furthermore, given that $P_A = 15MW$, company B achieves its highest profit by producing $P_B = 16MW$ AND given that $P_B = 16MW$, company A achieves its highest profit by producing $P_A = 15MW$.

(d) (ii) We can observe that under imperfect competition the market price and the generation profits are significantly higher than under perfect competition, since producers exercise market power and manipulate the prices to strategically increase their profits. As a result of the price elasticity of the demand, the increased price leads to lower demand (and consequently to lower generation levels).

(d) (iii) Each generating company is trying to maximize its profit which is equal to the difference between its revenue and its cost.

$$\Omega_A = P_A \cdot \pi - C_A(P_A) \tag{15}$$

$$\Omega_B = P_B \cdot \pi - C_B(P_B) \tag{16}$$

However, we cannot treat these two maximizations as separate optimization problems since they are links through the demand:

$$D = P_A + P_B \tag{17}$$

$$\pi = 125 - 1.7 \cdot D \tag{18}$$

We can around this difficulty by expressing the price as a function of the production of the two generators:

$$\pi = 125 - 1.7 \cdot (P_A + P_B) \tag{19}$$

The profit of the two companies than be expressed as:

$$\Omega_A = P_A(125 - 1.7P_A - 1.7P_B) - 35P_A - 0.37P_A^2 = -2.07P_A^2 - 1.7P_AP_B + 90P_A \tag{20}$$

$$\Omega_B = P_B(125 - 1.7P_A - 1.7P_B) - 39P_B - 0.25P_B^2 = -1.95P_B^2 - 1.7P_AP_B + 86P_B \tag{21}$$

The optimality condition for the maximization problem of company A is:

$$\frac{\partial \Omega_A}{\partial P_A} = -4.14P_A - 1.7P_B + 90 = 0 \tag{22}$$

The optimality condition for the maximization problem of company B is:

$$\frac{\partial \Omega_B}{\partial P_B} = -3.9P_B - 1.7P_A + 86 = 0 \tag{23}$$

(We only consider the partial derivative of the profit of a generator with respect to its own output because that is the only variable over which it has control)

Putting these optimality conditions in matrix form we get:

$$\begin{bmatrix} -4.14 & -1.7 \\ -1.7 & -3.9 \end{bmatrix} \begin{bmatrix} P_A \\ P_B \end{bmatrix} = \begin{bmatrix} -90 \\ -86 \end{bmatrix} \tag{24}$$

Solving this linear system we get the following productions for the NE:

$$P_A = 15.4496\ [MW] \tag{25}$$

$$P_B = 15.3168\ [MW] \tag{26}$$

Form which we can easily find:

$$\pi = 72.6970\ \left[\frac{£}{MWh}\right],\ D = 30.7664\ [MW],$$

$$\Omega_A = 494.0891\ [\tfrac{£}{h}],\ \Omega_B = 457.4808[\tfrac{£}{h}] \tag{27}$$

Solution to Question 3:

Part I:

(a)

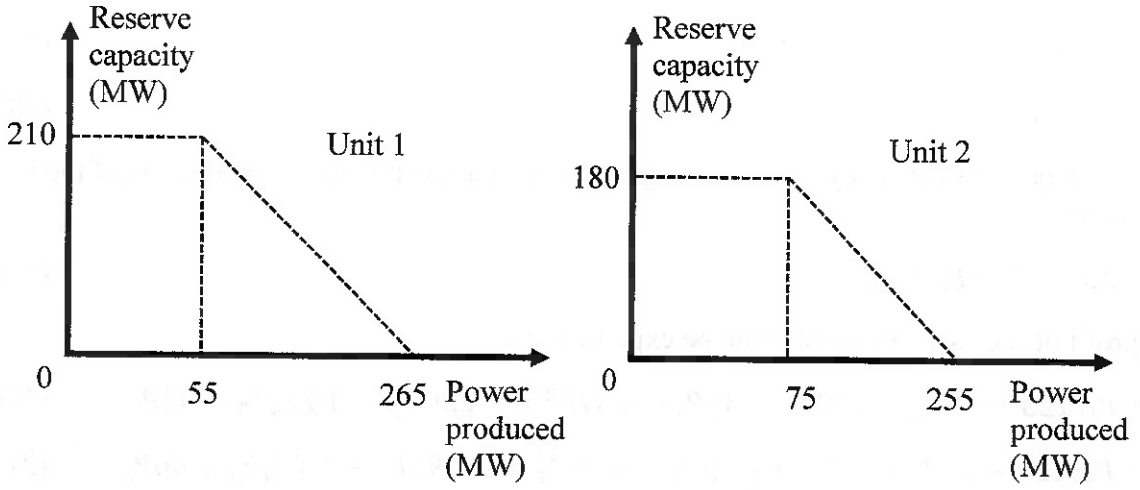


Figure 1. Amount of reserve that generating units 1 and 2 can provide as a function of the amount of electrical energy that they produce

(b) Given that the demand we consider is 420MW and that unit 4 has the lowest marginal operating cost and cannot provide reserve, we conclude immediately that this unit must produce its maximum output of 270MW for all values of the demand. Since units 1 and 2 are the only ones that can provide reserve and since unit 2 can provide at most 180MW, unit 1 has to provide at least 90MW. Given that unit 1 has a capacity of 265MW, its energy production must be less than 175 MW.

$0 \leq P_1 \leq 175$ (1)

Similarly, since unit 1 can provide at most 210MW of reserve, unit 2 has to provide at least 60MW. Its power output is thus limited to 195 MW:

$0 \leq P_2 \leq 195$ (2)

For the demand of 420MW, unit 2 is the marginal generator. It produces 150MW, that is, the power that is not supplied by unit 4, which runs at its full output of 270MW. Unit 1 does not produce energy, so it provides 210MW of reserve, which means unit 2 has to provide the remaining 60MW of reserve.

The marginal cost of unit 2 sets the price for energy at 19.5 £/MWh. At this demand, the inequality constraint for the minimum reserve requirement is not binding because units 1 and 2 provide more than enough reserve. The price of reserve is thus zero.

The market price for energy is equal to the bid price of unit 2. Since we assume that all generators bid at their marginal cost of production, this unit does not make an economic profit on the sale of energy. Given that the reserve price is zero, it does not make a profit on the provision of reserve either. In total, the profit that unit 2 derives from both the energy and reserve markets is zero.

(c) For a demand in the range between 465MW and 520MW, unit 1 is the marginal unit. In this demand range, unit 4 still runs at its full output of 270MW, the production of unit 2 is capped

at 195MW because it must provide at least 60MW of reserve. Unit 1 becomes the marginal generator and progressively increases its output from 0 to 55MW. For a demand which is greater than 520MW, unit 3 will become the marginal unit.

The price of energy is thus set at the marginal cost of unit 1, which is 24 £/MWh. To determine the price of reserve, we must figure out where an additional megawatt of reserve would come from and how much it would cost. Figure 1 in (a) shows that over this range of output, unit 1 provides 210MW of reserve, which is the maximum it is able to deliver under any circumstance. To get an additional megawatt of reserve beyond the basic 270 MW requirement, we would have to reduce the output of unit 2 by one megawatt. Instead of producing 195MW, it would thus produce only 194MW. To compensate for this reduction, we would have to increase the output of unit 1 by one megawatt. This extra megawatt from unit 1 would cost £24, while the reduction in the output of unit 2 would save £19.5. The net cost of getting an additional megawatt of reserve and hence the price of reserve is 24 – 19.5 = 4.5 £/MWh.

Even though unit 2 has a lower marginal cost than other units, its output is capped at 195MW by the reserve requirement constraint. Unit 1 is then marginal energy producer. The energy price is 24£/MWh, which means that unit 2 makes an 4.5£/MWh profit on each megawatt it produces. In this range of demand, unit 2 also make an 4.5£/MWh profit on every megawatt of reserve it provides. In total, the profit that unit 2 derives from both the energy and reserve markets is:

$$\text{Total profit of unit 2} = 195 \times 4.5 + 60 \times 4.5 = 1{,}147.5 \text{ £/h} \quad (3)$$

Part II:

(a) We apply the internal rate of return (IRR) method to estimate the profitability of the plant. To implement this method, we must determine the net cash flow for each year of this plant's lifetime. The cost of building the plant can be calculated as:

$$\text{Investment cost} = 1{,}010\text{£/kW} \times 450 \text{ MW} = \text{£}454{,}500{,}000 \quad (4)$$

Next, we need to estimate the annual production of this plant, given the utilization factor:

$$\text{Estimated annual production} = 450\text{MW} \times 0.85 \times 8{,}760\text{h/year} = 3{,}350{,}700\text{MWh} \quad (5)$$

Then calculate the annual cost of producing this energy:

$$\text{Annual production cost} = 3{,}350{,}700\text{MWh} \times 9{,}600\text{Btu/kWh} \times 1.15\text{£/MBtu} = \text{£}36{,}991{,}728 \quad (6)$$

$$\text{Annual revenue} = 3{,}350{,}700\text{MWh} \times 33\text{£/MWh} = \text{£}110{,}573{,}100\text{/year} \quad (7)$$

$$\text{Annual profit} = \text{£}110{,}573{,}100 - \text{£}36{,}991{,}728 = \text{£}73{,}581{,}372\text{/year} \quad (8)$$

Therefore, the total cash flow over the expected plant life (i.e. 30 years) is:

$$\text{Total cash flow} = 30\text{years} \times \text{£}73{,}581{,}372\text{/year} - \text{£}454{,}500{,}000 = \text{£}1{,}752{,}941{,}160 \quad (9)$$

(b) The IRR is a metric used to estimate the profitability of potential investments. The internal rate of return is a discount rate that makes the NPV of all cash flows from a particular project equal to zero.

$$NPV = 0 = CF_0 + \frac{CF_1}{(1+IRR)} + \frac{CF_2}{(1+IRR)^2} + \cdots + \frac{CF_n}{(1+IRR)^n} = \sum_{t=0}^{n} \frac{CF_t}{(1+IRR)^t} \quad (10)$$

Syldavia Power must then compare the value of IRR corresponding to the cash flow stream of the investment with their MARR before they make their decision. If IRR > MARR, then the investment is OK, if IRR < MARR, then the investment is not worth making. In this case, the IRR (16%) is lower than MARR (16.2%) which does not justify the investment in this plant.

(c)

Marginal cost = 2.4£/MBtu × 9600MBtu/kWh + 0.8£/MWh = 23.84£/MWh (11)

Annual loss = (23.84-23.55)£/MWh × 0.85 × 450MW × 8,760h/year = £971,703 (12)

Solutions to Question 4:

(a) ***Contracts for differences***: Producers and consumers of some commodities are sometimes obliged to trade solely through a centralized market. Since they are not allowed to enter into bilateral agreements, they do not have the option to use forward, future or option contracts to reduce their exposure to price risks. In such situations, parties often resort to *contracts for difference* that operate in parallel with the centralized market. In a contract for difference, the parties agree on a *strike price* and an amount of the commodity. They then take part in the centralized market like all other participants. Once trading on the centralized market is complete, the contract for difference is settled as follows:

• If the strike price agreed in the contract is higher than the centralized market price, the buyer pays the seller the difference between these two prices times the amount agreed in the contract.

• If the strike price is lower than the market price, the seller pays the buyer the difference between these two prices times the agreed amount.

A contract for difference thus insulates the parties from the price on the centralized market while allowing them to take part in this market. A contract for difference can be described as a combination of a call option and a put option with the same exercise price. Unless the market price is exactly equal to the strike price, one of these options will necessarily be exercised.

Financial Transmission Rights: Financial Transmission Rights deal with shortfalls in contracts for differences in the presence of network congestion. They are defined between any two nodes in the network and entitle their holders to a revenue equal to the product of the amount of transmission rights bought and the price differential between the two nodes. This amount is exactly what is needed to ensure that a contract for difference can be settled. In conclusion, FTRs completely isolate their holders from the risk associated with congestion in the transmission network. They provide a perfect hedge against variations in nodal prices.

(b) Given that the power on the line flows from Borduria to Syldavia, it holds:

$$P_{BP} + P_{BG} = D_B + F \quad (1)$$

$$P_{SE} + P_{SS} = D_S - F \quad (2)$$

Given that the line is congested, it holds:

$$F = 400\ [MW] \quad (3)$$

Given (1), (3) and the demand data, it holds:

$$P_{BP} + P_{BG} = 900 \quad (4)$$

Furthermore, the marginal cost of BP and BG is the same since they are located at the same bus:

$$\frac{\partial C_{BP}(P_{BP})}{\partial P_{BP}} = \frac{\partial C_{BG}(P_{BG})}{\partial P_{BG}} \rightarrow 10 + 0.01 * P_{BP} = 16 + 0.01 * P_{BG} \quad (5)$$

The system of (4) and (5) yields:

$$P_{BP} = 750\ [MW] \tag{6}$$

$$P_{BG} = 150\ [MW] \tag{7}$$

However, since $P_{BP}^{max} = 600\ [MW]$, the final dispatch for the generators of Borduria is:

$$P'_{BP} = 600\ [MW] \tag{8}$$

$$P'_{BG} = 300\ [MW] \tag{9}$$

Since BP is constrained by its maximum output limit, the locational marginal price in Borduria is given by the marginal cost of BG:

$$\pi_B = 16 + 0.01 * P'_{BG} = 19\ [\tfrac{£}{MWh}] \tag{10}$$

Given (2), (3) and the demand data, it holds:

$$P_{SE} + P_{SS} = 1100 \tag{11}$$

Furthermore, the marginal cost of SE and SS is the same since they are located at the same bus:

$$\frac{\partial C_{SE}(P_{SE})}{\partial P_{SE}} = \frac{\partial C_{SS}(P_{SS})}{\partial P_{SS}} \rightarrow 13 + 0.02 * P_{SE} = 23 + 0.02 * P_{SS} \tag{12}$$

The system of (11) and (12) yields:

$$P_{SE} = 800\ [MW] \tag{13}$$

$$P_{SS} = 300\ [MW] \tag{14}$$

However, since $P_{SE}^{max} = 500\ [MW]$, the final dispatch for the generators of Syldavia is:

$$P'_{SE} = 500\ [MW] \tag{15}$$

$$P'_{SS} = 600\ [MW] \tag{16}$$

Since SE is constrained by its maximum output limit, the locational marginal price in Syldavia is given by the marginal cost of SS:

$$\pi_S = 23 + 0.02 * P'_{SS} = 35\ [£/MWh] \tag{17}$$

The cost of each of the generators is calculated by replacing their respective outputs (8), (9), (15), (16) in the respective given cost functions. The revenue of each of the generators is calculated by multiplying their outputs with the locational marginal price at the bus where they are connected to. The profit of each of the generators is calculated by subtracting their respective costs from their respective revenues. The locational marginal prices (in £/MWh), optimal dispatches (in MW), and these quantities of costs, revenues, profits (in £/h) are presented in the next table:

	Price	Amount	Cost	Revenue	Profit
Borduria Power	19	600	7,800	11,400	3,600
Borduria Gen	19	300	5,250	5,700	450
Syldavia Energy	35	500	9,000	17,500	8,500
Syldavia Supply	35	600	17,400	21,000	3,600

The charges of each of the two retail companies is calculated by multiplying the demand that they serve with the locational marginal price at the bus where this demand is located. These charges are equal to 19£/MWh × 500MW = 9,500£/h for Borduria Retail and 35£/MWh × 1,500MW = 52,500£/h for Syldavia Retail.

The congestion surplus is equal to the difference between the total charges of the two retail companies and the total revenues of the four generators, which is 9,500 + 52,500 – (11,400 + 5,700 + 17,500 + 21,000) = 6400£/h.

(c) Based on the definition of the contract for differences given in (a), the revenue for the seller of each of the given contracts (which is equal to the charge for the buyer of the respective contract) is given in the following table:

Seller	**Buyer**	**Amount [MWh]**	**Strike Price [£/MWh]**	**Delivery Point**	**Seller's revenue and buyer's charge [£/h]**
Borduria Power	Borduria Gen	200	19	Borduria	(19-π_B)*200=0
Borduria Power	Syldavia Retail	100	19	Borduria	(19-π_B)*100=0
Borduria Power	Borduria Retail	300	19	Borduria	(19-π_B)*300=0
Borduria Gen	Borduria Retail	300	21	Borduria	(21-π_B)*300 =600
Borduria Gen	Syldavia Energy	100	33	Syldavia	(33-π_S)*100 =-200
Borduria Gen	Syldavia Retail	100	34	Syldavia	(34-π_S)*100 =-100
Borduria Retail	Syldavia Retail	100	35	Syldavia	(35-π_S)*100=0
Syldavia Energy	Syldavia Retail	600	37	Syldavia	(37-π_S)*600 =1,200
Syldavia Supply	Syldavia Retail	600	38	Syldavia	(38-π_S)*600 =1,800

Given the revenues and charges associated with the above contracts and the revenues, costs and charges calculated in (b), the sell/buy prices (in £/MWh), sell/buy amount (in MW), and net costs, revenues and profits (in £/h) for the four generators are:

	Sell price	Sell amount	Buy price	Buy amount	Cost	Revenue	Profit
Borduria Power	19	200/100/300	-	-	7,800	11,400	3,600
Borduria Gen	21/33/34	300/100/100	19	200	5,250	6,000	750
Syldavia Energy	37	600	33	100	9,000	18,900	9,900
Syldavia Supply	38	600	-	-	17,400	22,800	5,400

And the net charges for the two retail companies are 9,500 + 600 = 10,100£/h for Borduria Retail and 52,500 – 100 + 1,200 +1,800 = 55,400£/h for Syldavia Retail.

The total income of the ISO is equal to the difference between the total charges of the two retail companies and the total revenues of the four generators and is again equal to the congestion surplus, i.e. 10,100 + 55,400 – (11,400 + 6,000 + 18,900 + 22,800) = 6,400£/h.

(d) Based on the definition of the Financial Transmission Rights given in (a), the revenue for the holder of each of those rights is given in the following table:

Participant	Amount of FTR purchased [MW]	Revenue of FTR holder
Borduria Gen	200	$200 * (\pi_S - \pi_B) = 3200$
Borduria Retail	100	$100 * (\pi_S - \pi_B) = 1600$
Syldavia Retail	100	$100 * (\pi_S - \pi_B) = 1600$

Given the revenues associated with the above FTR, the revenues and charges associated with the contracts for differences calculated in (c) and the revenues, costs and charges calculated in (b), the sell/buy prices (in £/MWh), sell/buy amount (in MW), and net costs, revenues and profits (in £/h) for the four generators are:

	Sell price	Sell amount	Buy price	Buy amount	Cost	Revenue	Profit
Borduria Power	19	200/100/300	-	-	7,800	11,400	3,600
Borduria Gen	21/33/34	300/100/100	19	200	5,250	9,200	3,950
Syldavia Energy	37	600	33	100	9,000	18,900	9,900
Syldavia Supply	38	600	-	-	17,400	22,800	5,400

And the net charges for the two retail companies are 10,100 – 1,600 = 8,500£/h for Borduria Retail and 55,400 – 1,600 = 53,800£/h for Syldavia Retail.

The total income of the ISO is equal to the difference between the total charges of the two retail companies (8,500 + 53,800 = 62,300£/h) and the total revenues of the four generators (11,400 + 9,200 + 18,900 + 22,800 = 62,300£/h) and is now equal to zero, as it has used the congestion surplus to pay the holders of the FTR.